Assignment 6: Implementation of Data Integrity Constraints

COLLEGE MANAGEMENT SYSTEM

DBMS Assignment 6

Database: CollegeManagementSystem

SQL Platform: MySQL 8.0.16 or later

Project scope: departments, faculty, students, courses, and enrollments.

Note: Query outputs described in this report are expected results from the included fictional sample data. Run the SQL script in MySQL Workbench and capture actual screenshots if your instructor requires them.

Purpose

Constraints prevent invalid, duplicate, or orphaned records and preserve reliable relationships.

Constraint explanations

PRIMARY KEY uniquely identifies each row. FOREIGN KEY ensures referenced departments, faculty, students, and courses exist. UNIQUE prevents duplicate department names, emails, register numbers, course codes, and duplicate student-course-term enrollments. CHECK validates non-negative salaries, course credits from 1–6, semester from 1–8, allowed employment status, and allowed grades. DEFAULT supplies Active status or NA grade when no value is provided.

Validation SQL

```
-- Expected to fail: duplicate department name
INSERT INTO Departments (department_name,office_location)
VALUES ('Commerce','Another Office');

-- Expected to fail: invalid credits
INSERT INTO Courses (course_code,course_name,credits,department_id,faculty_id)
VALUES ('CS999','Invalid Course',9,1,1);

-- Expected to fail: foreign key department 999 does not exist
INSERT INTO Students (register_no,student_name,email,admission_year,department_id)
VALUES ('TEST001','Test Student','test@college.edu',2026,999);

-- Inspect definitions and relationship data
SHOW CREATE TABLE Enrollments;
SELECT s.student_name,d.department_name
FROM Students s JOIN Departments d ON s.department_id=d.department_id;
SELECT COUNT(*) AS student_count FROM Students;
```

Expected behavior

Valid data is accepted. Duplicate unique values, invalid CHECK values, and nonexistent foreign-key references are rejected. Run negative tests individually in a practice database.

End of Assignment Report